

AI for Bharat Hackathon

Powered by **aws**



Team Name : Velocity Tech...

Team Leader Name : Pranay Kumar Karvi

Problem Statement : 4. [Student Track] AI for Learning & Developer Productivity

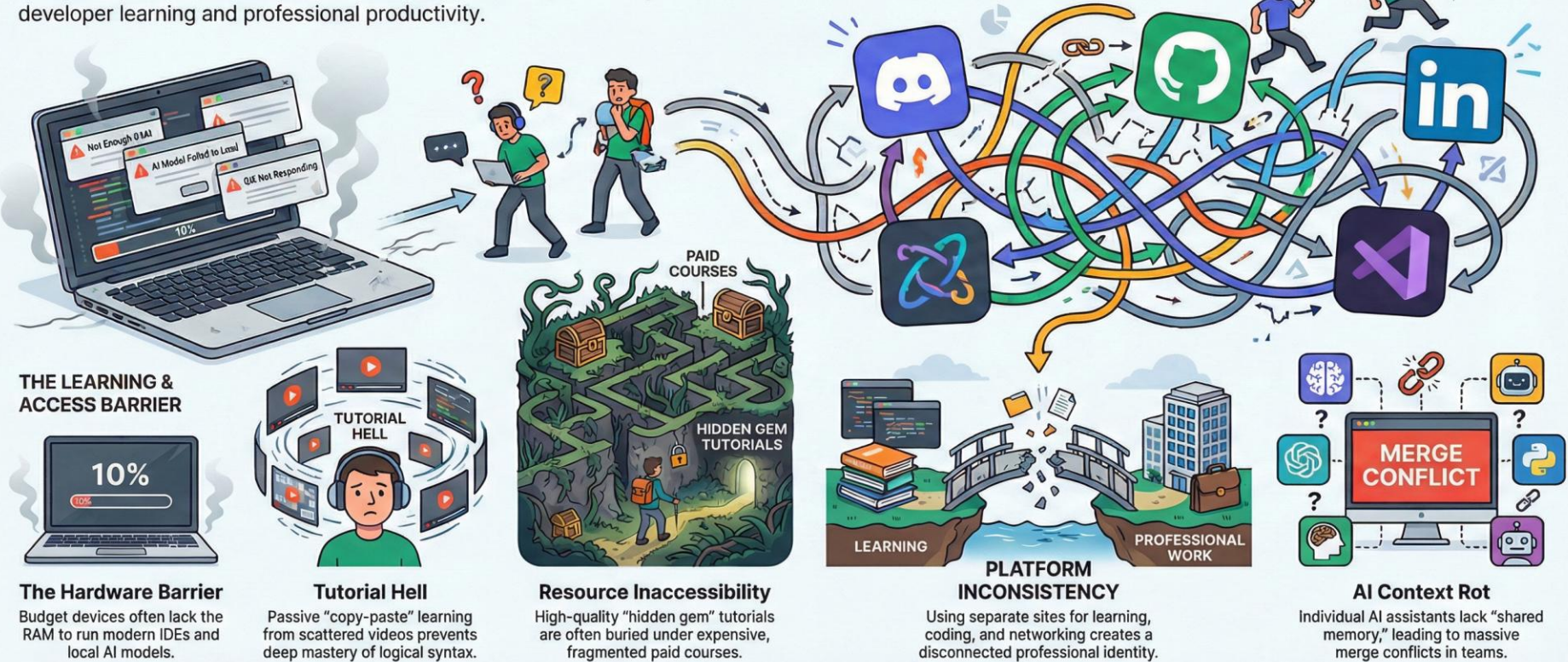
The Developer's Deadlock: Visualizing the Fragmentation Tax

To illustrate the core barriers and inefficiencies currently hindering developer learning and professional productivity.

THE PRODUCTIVITY & PLATFORM CRISIS

Cognitive Context-Switching

Constant context-switching between Discord, GitHub, and LinkedIn drains cognitive energy and time.



Velocity: The Multi-Agent Engineering Ecosystem

THE TRIPLE-THREAT STRATEGY (PLATFORMS)

VELOCITY WEB (THE NOVICE ENTRYWAY)



A zero-install browser IDE using AWS AppStream 2.0 to turn budget laptops into workstations.

VELOCITY DESKTOP (THE PRO POWERHOUSE)



A native IDE that offloads multi-agent orchestration to the cloud to maintain local speed.

VELOCITY MOBILE (THE COMMAND CENTER)



A companion app for remote "Save Gate" approvals and managing the Verified Skill Wallet.

PLATFORM COMPARISON

	VELOCITY WEB	VELOCITY DESKTOP	VELOCITY MOBILE
PRIMARY TECH STACK:	AWS AppStream 2.0	Amazon Bedrock	AWS Nitro Enclaves
CORE USER BENEFIT:	Zero-install accessibility for budget hardware.	High-velocity parallel AI orchestration for pros.	Immutable proof of talent via Skill Hashes.

CORE TECHNICAL INNOVATIONS



THE CONTEXT JANITOR

A background service maintaining a "Source of Truth" in a Vector Database to sync agents.



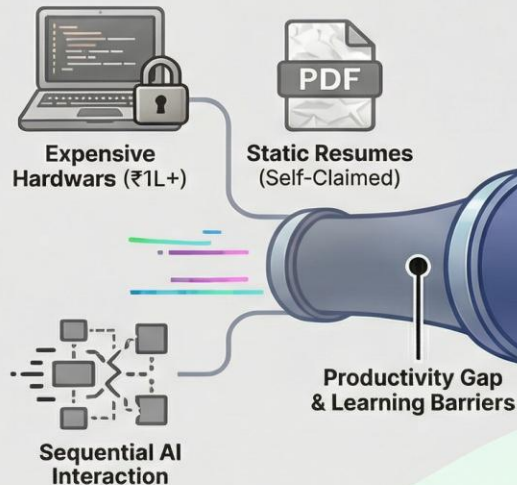
VERIFIED SKILL WALLET

Replaces resumes with immutable "Skill Hashes" and AI narrated playbacks verified by AWS Nitro Enclaves.



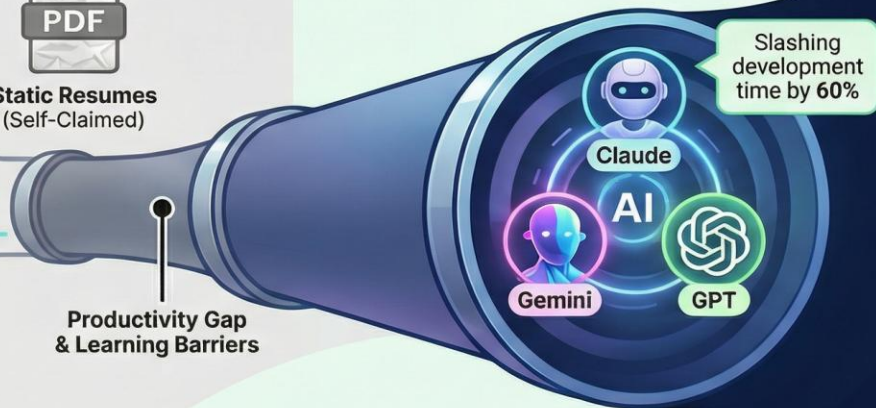
Velocity: The Future of High-Velocity Engineering & Learning

TRADITIONAL DEVELOPMENT (FRAGMENTED & SLOW)



VELOCITY ECOSYSTEM (ACCELERATED & VERIFIED)

Parallel Multi-Agent Engine



COMPARISON: TRADITIONAL vs. VELOCITY

Feature	Traditional IDE	Velocity Ecosystem
Hardware Req.	 Expensive (₹1L+)	 Budget (₹15k Chromebook)
AI Interaction	 Sequential/Single	 Parallel/Multi-Agent
Talent Proof	 Self-Claimed Resume	 Verified 'Skill Wallet'



Budget Hardware
(₹15k Chromebook) to
16GB Cloud Workstation

AI-Smart Roadmaps & Skill Gates



Verified Skill Wallet

Replaces unreliable PDFs with
immutable 'Skill Hashes' &
AI-narrated coding playbacks



The Context Janitor

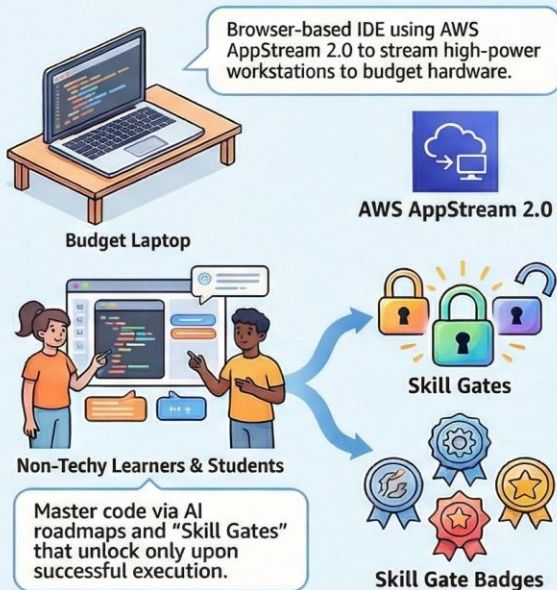
Maintains 'Source of Truth',
preventing code conflicts

Model-Agnostic Sovereignty:

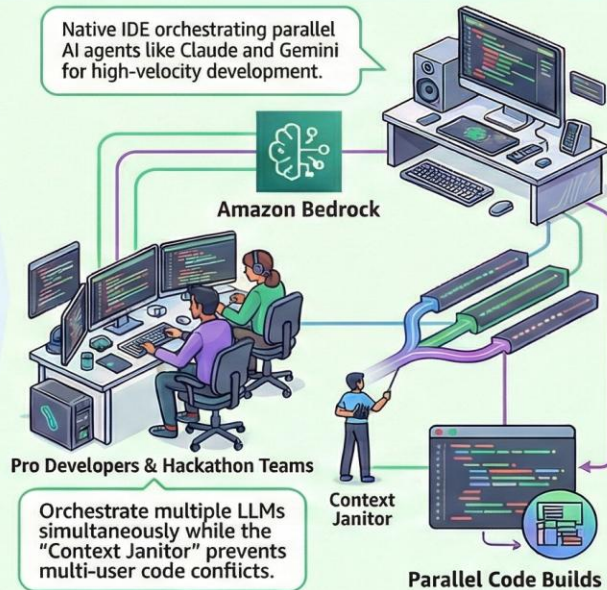
Choose your 'Dream Team' of LLMs

Velocity: The Multi-Agent Engineering Ecosystem

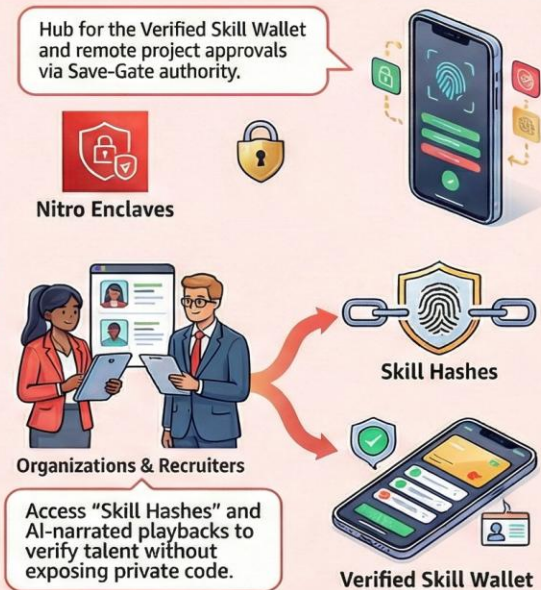
Velocity Web: The Novice Entryway



Velocity Desktop: The Pro Powerhouse



Velocity Mobile: The Career Command Center



Accessibility & Learning



AppStream 2.0



Skill Gate Badges

High-Speed Production



Amazon Bedrock



Parallel Code Builds

Career & Team Ops



Nitro Enclaves



Verified Skill Wallet

Velocity: The Multi-Agent Engineering Workflow

Triple-Threat Access Points (User Input)



Velocity Web
AWS AppStream 2.0
streaming to browser



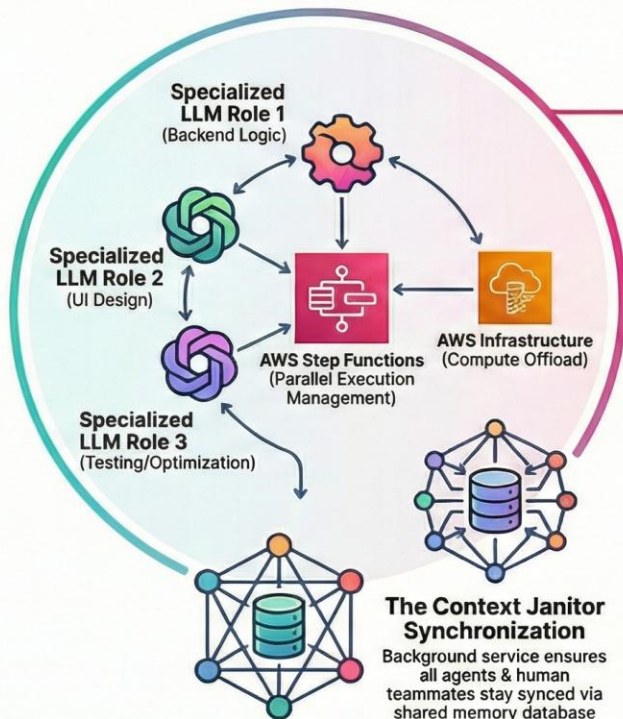
Velocity Desktop
Pro Powerhouse: Native IDE
offloading AI to Amazon Bedrock



Velocity Mobile
Command Center: Career hub and
remote "Save-Gate" approval

THE ENGINEERING CORE (INPUT TO EXECUTION)

User-Defined Parallel Orchestration



THE VERIFICATION LOOP (EXECUTION TO CAREER)

**60% REDUCTION IN
DEVELOPMENT TIME**

Velocity (Simultaneous Processing)

Standard (One-Request-at-a-Time Bottleneck)

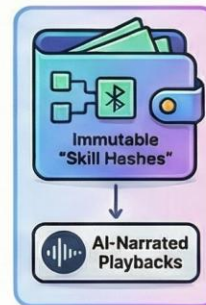
Simultaneous processing of backend logic and UI design eliminates the standard one-request-at-a-time AI bottleneck.



AWS Nitro Enclaves
(Secure Processing)

Privacy-First Behavioral Profiling

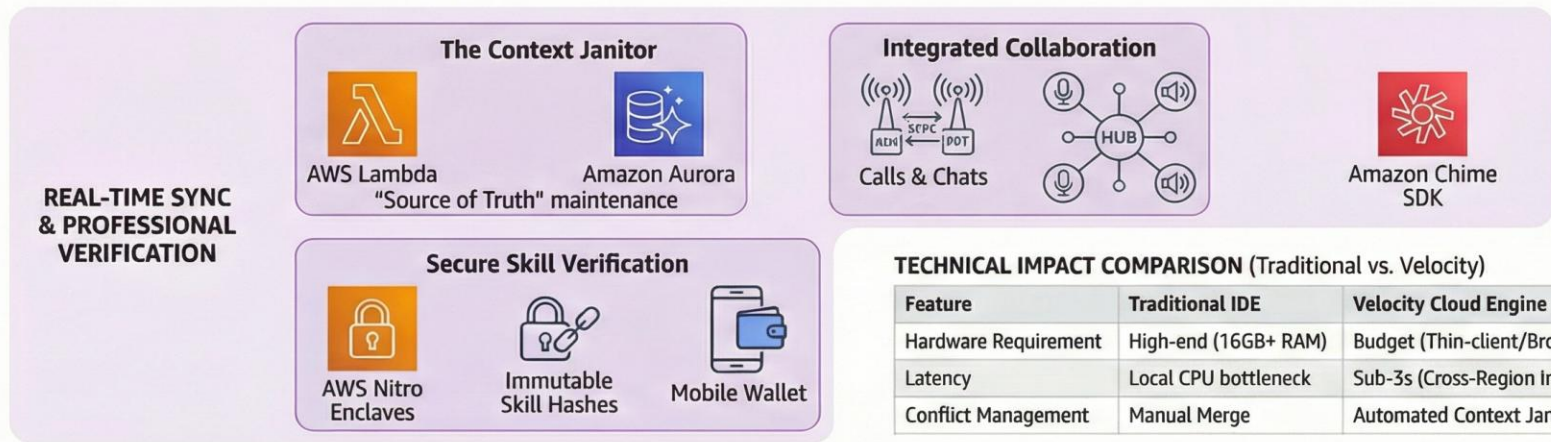
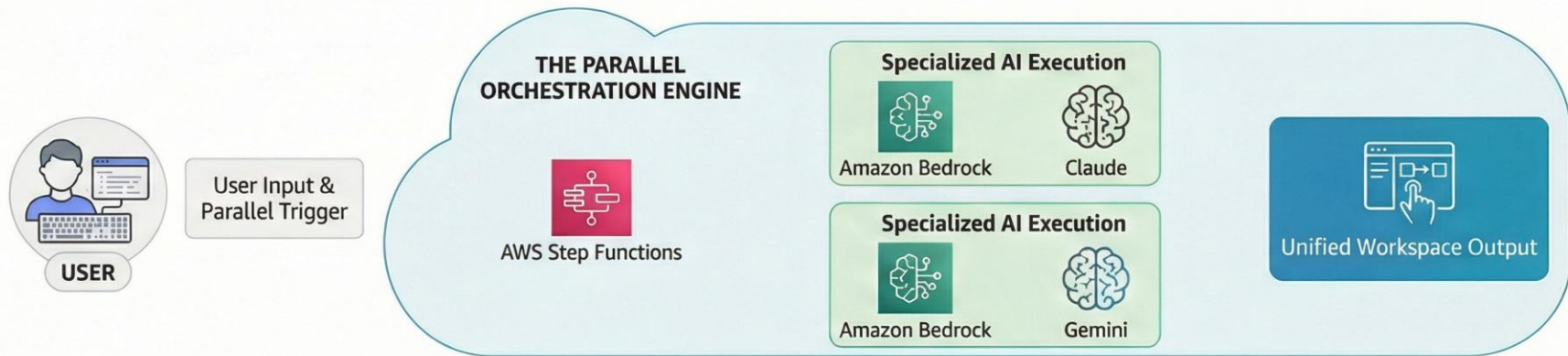
Coding logs processed in
a secure "black box" to
prove expertise



The Meritocracy Loop

Recruiters watch
30-second verified clips
of problem-solving to
confirm authentic
technical talent.

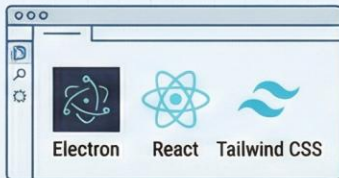
Velocity: The High-Velocity Cloud Architecture



VELOCITY: THE MULTI-AGENT TECHNICAL BLUEPRINT

A model-agnostic platform offloading compute-intensive tasks to AWS, orchestrating multiple LLMs in parallel, and verifying expertise via secure enclaves.

MULTI-PLATFORM & COLLABORATION STACK



Frontend & Client Frameworks
Built using Electron, React, and Tailwind CSS for cross-platform IDE performance.



Amazon Chime SDK
Provides low-latency, integrated voice and video calls directly within the IDE workspace.



AWS AppStream 2.0 Streaming



Streams high-performance virtual workstations to any browser, enabling 16GB RAM cloud environments.

MULTI-AGENT & SECURITY ENGINE



Amazon Bedrock



Claude



Gemini



GPT

Model-Agnostic LLM Orchestration
Leverages Amazon Bedrock to support user-selected models including Claude, Gemini, and GPT.



Behavioral Logs



Privacy-First Verification (Nitro Enclaves)
Securely processes behavioral logs to generate immutable Skill Hashes without exposing code.



Immutable Skill Hashes



AWS Step Functions



Logic



UI

Parallel Execution via Step Functions
Triggers specialized AI agents simultaneously to build logic and UI in sync.

COMPREHENSIVE TECHNOLOGIES & SERVICES

COMPUTE & AI	DATA & STORAGE	SECURITY & SDKS
Amazon Bedrock	Amazon Aurora (Vector Search)	AWS Nitro Enclaves
AWS Lambda	Redis	Amazon Chime SDK
AWS Step Functions	MongoDB	Amazon Kendra
LLMs (Claude, Gemini, GPT)	Amazon Personalize	AppSync

Velocity: The ROI of Next-Gen Engineering

Velocity is a model-agnostic, cloud-powered engineering ecosystem that democratizes high-and development. By offloading compute to AWS, it allows users to learn and build on budget hardware while earning AI-verified credentials that unlock global hiring opportunities.



The Thin Client

IMPLEMENTATION & ENTRY COSTS

**₹2,000 (\$24 USD)
Monthly Subscription**

Affordable access to high-performance IDE features and cloud-streamed compute power.



Zero Hardware Upgrade Costs

Thin-client architecture runs pro-grade AI agents on budget ₹20,000 laptops.



Roadmap Generation Tokens

Small per-use fees for AI-generated, customized learning paths using free web resources.

EARNINGS & ROI POTENTIAL

The "Coupon Loop" Incentives

Users earn 5-10% subscription discounts by hitting 30-minute active learning milestones.



The Skill Wallet

10x Career Valuation

AI-Verified Skill Wallets prove talent to global recruiters, bypassing traditional resume barriers.



COMPARATIVE VALUE EXCHANGE

Category	Cost (INR / USD)	Potential Return (INR / USD)
Standard Access	₹2,000 / ~\$24	10% Discount via "Coupon Loop"
Skill Validation	Small Token Fee	Access to Global \$100k+ Engineering Roles
Team Infrastructure	AWS Usage Rates	60% Reduction in Dev-Time (via Parallel Agents)

Recruitment Matchmaking Revenue



Organizations pay for access to high-fidelity, behavioral-based hiring signals and verified playbacks.

Innovation partner **I12S**
HACKZESTLE

Media partner **YOURSTORY**

AI for Bharat Hackathon

Powered by **aws**

Thank You

